Shri G.S. Institute of Technology and Science
Department of Applied Mathematics and Computational Science
B.Tech I Year (July-Dec.2025) Group-B
Assignment I: MA 10509 (Mathematics for Data Science)

Last Date of Submission: 30/09/2025

	Questions	CO
Q.1	(i) Identify and elaborate on key Indian contributions to the computational and modern era of data science. (ii) Explain the post-independence growth of statistics and data science in India. (iii) Compare the Contributions of India in Statistics during ancient and modern eras.	1
Q.2	(i) A proof reads through 73 pages manuscript. The number of mistakes found on each of the pages are summarized in the table below Determine the mean number of mistakes found per page	1

No. of Mistakes	1	2	3	4	5	6	7
No. of Pages	5	9	12	17	14	10	6

(ii) Calculate the median and standard deviation for the following table, given the age distribution of 542 members.

Age in Years	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 – 80	80 - 90
No. of members	3	61	132	153	140	51	2

Q.3	**(i)** Find the mean, median, mode and standard deviation for the following data							1

Size of item	6	7	8	9	10	11	12
Frequency	3	6	9	13	8	5	4

(ii) In a moderately asymmetrical frequency distribution, the values of median and arithmetic mean are 72 and 78 respectively; estimate the value of the mode.

Q.4	**(i)** The following data relates to the heights of 10 students (in cms) in a school. Calculate the range and coefficient of range. 158, 164, 168, 170, 142, 160, 154, 174, 159, 146. **(ii)** The following are the weights of 10 children admitted in a hospital on a particular day. Find the mean deviation about mean, median and their coefficients of mean deviation. 7,4,10,9,15,12,7,9,9,18.	1
Q.5	Find the mean deviation from the median and mode from the following data.	1

x	56	63	70	77	84	91	98
f	3	6	14	16	13	6	2

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.6	**(i)** A bag contains 6 white, 8 red and 5 black balls. A ball is drawn at random. Find the probability that it is white. **(ii)** For any two events A and B, given that $P(A) = 0.4, P(B) = 0.7$ *and* $P(A \cap B) = 0.3$. Then find $P(A \cup B)$.	2
Q.7	(i) A card is drawn from a well shuffled pack of 52 cards. Find the probability that it is either a spade or a king. (ii) If A and B are two events, where P(A) = ½ , P(B)=⅓ and P(A∩B) =¼. Then calculate (i) P(A|B) (ii) P(B|A) (iii) P(A ∪ B) (iv) P(A ∪ B), when A and B are independent.	2
Q.8	(i) Probability that A speaks the truth is 3/5 and B speaks the truth is 5/8. In what percentage of cases they are likely to contradict each other? (ii) 5 man out of 100 and 25 women out of 1000 are colour blind. A colour blind person is chosen at random. What is the probability of his being a male? (Assume that male and female are in equal proportion).	2
Q.9	(i) Eight coins are thrown simultaneously. Find the probability of obtaining at least six heads. (ii) Assuming the probability of male birth as 1/2, find the probability that a family of three children will have: (i) at least one girl, (ii) two boys and a girl, (iii) at the most two girls.	2
Q.10	(i) If 2% of the electric bulbs manufactured by a certain company are defective, find the probability that in a sample of 200 bulbs (i) less than two bulbs are defective (ii) more than two bulbs are defectives. (e^{-4} =0.0183) (ii) Suppose that a manufactured product has 2 defects per unit of product inspected. Using Poisson's distribution, calculate the probability of finding a product without any defect, 3 defects and 4 defects. (e^{-2}=0.718 0.135)	2

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